

FELIX BLANCHARD

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GPA: 7.5 | [LinkedIn](#) | [Portfolio](#)

SUMMARY

I am a responsible and versatile Bachelor of Software Engineering (Hons) student at the University of Canterbury who values honesty, reliability and efficiency. I am always eager to learn new skills and excel in collaborative environments.

EDUCATION

Bachelor of Engineering (Hons), University of Canterbury

Feb 2023 - Nov 2026

Major: Software Engineering

WORK EXPERIENCE

Software Engineering Intern, Allied Telesis

Nov 2025 - Feb 2026

- Contributed to a software team, conducting technical research into location tracking protocols, analysing and extracting key data from academic papers, to assist project development.
- Developed embedded C programs for networking hardware, including access points and switches.
- Compiled and ran builds for hardware testing before documenting findings for future reference by the software team.

Private Tutor - Coding

Jan 2024 - May 2025

- Privately tutored a first year computer science course, COSC131.

Geotechnical field worker, Deane Consultancy Limited

Nov 2023 - Feb 2024

- Conducted borehole sampling, shear vane testing and soil analyses, to be used in technical reports.
 - Gained insight into the operation of a professional geotechnical engineering firm.
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OVERVIEW OF SOFTWARE PROJECTS

LLM Benchmarking

2026 – Present

- Investigating novel reinforcement learning techniques for training large language models (Qwen3) on domain specific tasks. This bulk of the project focuses on evaluating model performance and training efficiency to determine the viability of these techniques for future implementation.

Location Tracking – FTM, BlueZ

2025 – 2026

Investigated various techniques of performing location tracking in an urban environment:

- **WiFi RSSI** - Programmed an ESP32 to scan nearby access points and then wrote a python script to parse, store, plot and finally estimate its own position using several trilateration techniques (linear vs nonlinear least squares).
 - **WiFi RTT** - Programmed an ESP32 to perform FTM (Fine time measurement) a IEEE outlined technique that utilises wifi round trip time to measure a device's distance from an access point. The bulk of this project was combing through ESP-IDF's documentation before settling on utilising its wifi library.
 - **BLE RSSI** - Added support for BLE scans to proprietary devices. Scans then propagated select information across network pipelines to designated collector devices for location estimation. This allows for servers to estimate the positions of nearby Bluetooth Low Energy devices.
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Renovation Web App

2025

- Full stack development in a team of 7 to create a renovation themed social media/renovation manager.

RIP v2 Routing Daemon

2025

- Developed a Python implementation of the RIP v2 Protocol using UDP sockets.
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ADDITIONAL QUALIFICATIONS/ AWARDS

University of Canterbury Hiranga Scholarship (top-achieving students) 2023

Mount Aspiring College Upper Clutha History Prize 2022

Gold Duke of Edinburgh Award 2022

Grade 3 Piano | Trinity College London 2018

NCEA Levels 1–3 with Excellence Endorsement (2020–2022)

COMMUNITY INVOLVEMENT

Events Manager, Climbing Club

Jan 2025 - Nov 2025

University of Canterbury UCSA Club

- Organised, managed and led club events.
- Thoroughly planned club trips to mitigate the risks of outdoor climbing, especially when introducing new members to the sport.

Shop Worker, The Salvation Army

Feb 2020 - Feb 2022

- Volunteered at the Devonport Salvation Army store two times a week during school.

Grocery Deliveries, Student Volunteer Army

2020 - 2021

- I delivered groceries for the elderly and immobile during Auckland's covid lockdowns.

I'm an active member of several clubs: EngReview, UC³ (Climbing Club), CompSoc (Computer Society). I am a very outgoing person and love to throw myself into clubs and their communities, as I have found it is a great way to meet and connect with a variety of likeminded people who share my interests.

SKILLS AND STRENGTHS

Languages:

Python, Java, Javascript/Typescript, PHP, C, C++, SQL, Markdown

Framework & Libraries:

Spring Boot, ESP-IDF, FreeRTOS, React, PyTorch, Gradle, MediaPipe, OpenCV, TRL, Node.Js

Systems & Software:

Linux (Bash), Git, Jira, Figma, SketchUp, GIMP, Microsoft365

OTHER INTERESTS

Sport climbing, surfing, skiing, hiking, hockey, music, chess, Project Euler, cricket

REFEREE

Scott Parlene | Senior Software Engineer | Allied Telesis | Scott.Parlane@alliedtelesis.co.nz

Scott was my manager during my Allied Telesis Summer Internship.
